

TeleBill - Telehealth Billing & Insurance Claim Submission Portal

1. Introduction

TeleBill is a web-based **telehealth billing & claims submission** system for clinics, group practices, and virtual-care networks. It streamlines **charge capture** from telehealth encounters, **coding (CPT/HCPCS, ICD-10-CM)**, **modifiers & POS**, **claim creation (837P ref)**, **payer rules & edits (claim scrubbing)**, **prior authorization references**, **EOB/ERA posting (835 ref)**, **patient balance & statements**, and **denial management**.

Phase-1 focuses on **professional claims** (CMS-1500 / 837P reference) for **telehealth** encounters, with in-app notifications only. External clearinghouses, payer APIs, and e-eligibility/EVV are modeled as **references** (payloads stored, not transmitted).

Technology: REST API backend (Spring Boot / .NET Core), Angular/React frontend, RDBMS (MySQL/PostgreSQL/SQL Server).

Actors / Users

- **Front Desk / Billing Specialist** – Capture charges, verify demographics & coverage, submit claims.
- **Coder** – Code encounters, add modifiers, ensure compliance with payer policy.
- **Provider** – Review charges, attest clinical documentation.
- **Payer Ops / AR Analyst** – Work denials, post EOB/ERA (ref), follow-ups.
- **Finance** – Review statements, collections references, write-offs (ref).
- **Admin** – Maintain fee schedules, payer setup, rules/edits, users & roles.

2. Module Overview

2.1 Identity & Access Management – Auth, RBAC, audit logs

2.2 Provider, Payer & Fee Schedule Master – Providers, payers, payer plans, fees

2.3 Patient, Coverage & Eligibility (Reference) – Demographics, coverage, eligibility ref

2.4 Telehealth Encounter Intake & Charge Capture – Encounter import, charge lines, attestations

2.5 Coding, Modifiers & POS – CPT/HCPCS, ICD-10-CM, modifiers, POS 02/10

2.6 Claim Building (837P Ref) & Scrubbing Rules – Claim assembly, payer edits, error queue

2.7 Prior Authorization & Attachments (Reference) – PA tracking, supporting docs

2.8 Submission, Clearinghouse & Payer ACK (Reference) – Batch submission refs, acks

2.9 ERA/EOB Posting, Patient Balance & Statements (Reference) – Remits, balances, statements

2.10 Denials, Appeals & AR Worklists – Denial buckets, appeals, follow-ups

2.11 Reports, Dashboards & Audit – KPIs, exports, immutable audit

2.12 Notifications & Alerts – Scrub errors, ACK/Remit refs, denials, statement runs

3. Architecture Overview

- **Frontend:** Angular/React SPA
- **Backend:** REST microservices (Identity, MasterData, Encounters/Charges, Coding, Claims, Submissions, AR/Denials, Reports, Notifications)
- **Database:** MySQL/PostgreSQL/SQL Server
- **Integrations (Phase-1):** Clearinghouse/payer submission (837), remittance (835), e-eligibility (270/271), provider NPI/PECOS – **references only** (payloads stored; no transmission)
- **Security:** RBAC, TLS, encryption at rest (configurable), immutable audit logs, PHI minimization
- **Scalability:** Stateless services, job schedulers for **scrub batches, submission packs, statement runs, AR worklists**

4. Module-Wise Design

4.1 Identity & Access Management Module

Features:

- Login, logout, password reset, lock/unlock
- Role-based access (FrontDesk/Coder/Provider/AR/Finance/Admin)
- Session/token lifecycle, inactivity timeouts
- Immutable audit logging for read/write events
- User provisioning, deactivation, role mapping
- Site/organization scoped access

Entities:

- **User**
 - UserID

- Name
- Role
- Email
- Phone
- Status
- **AuditLog**
 - AuditID
 - UserID
 - Action
 - Resource
 - Timestamp
 - Metadata

4.2 Provider, Payer & Fee Schedule Master Module

Features:

- Provider registry (NPI, taxonomy, telehealth enrollment flags)
- Payer & plan master (payer IDs, clearinghouse IDs)
- Fee schedule setup (CPT/HCPCS → allowed/contracted rates)
- Telehealth POS/modifier defaults per payer plan
- Effective dating and status lifecycle

Entities:

- **Provider**
 - ProviderID
 - NPI
 - Name
 - Taxonomy
 - TelehealthEnrolled (Yes/No)
 - ContactInfo
 - Status
- **Payer**
 - PayerID

- Name
- PayerCode
- ClearinghouseCode
- ContactInfo
- Status
- **PayerPlan**
 - PlanID
 - PayerID
 - PlanName
 - NetworkType (HMO/PPO/Medicare/Medicaid/Commercial)
 - POSDefault (02/10)
 - TelehealthModifiersJSON
 - Status
- **FeeSchedule**
 - FeeID
 - PlanID
 - CPT_HCPCS
 - ModifierCombo
 - AllowedAmount
 - EffectiveFrom
 - EffectiveTo
 - Status

4.3 Patient, Coverage & Eligibility Module (Reference)

Features:

- Patient demographics; guarantor info
- Coverage capture (primary/secondary/tertiary)
- Eligibility request/response **reference** (store payload/decision only)
- Insurance card images & coverage notes

Entities:

- **Patient**

- PatientID
- MRN
- Name
- DOB
- Gender
- AddressJSON
- ContactInfo
- Status
- **Coverage**
 - CoverageID
 - PatientID
 - PlanID
 - MemberID
 - GroupNumber
 - EffectiveFrom
 - EffectiveTo
 - Status
- **EligibilityRef**
 - EligibilityID
 - CoverageID
 - RequestPayloadURI
 - ResponsePayloadURI
 - CheckedDate
 - Result (Eligible/Ineligible/Partial)
 - Notes

4.4 Telehealth Encounter Intake & Charge Capture Module

Features:

- Import or register telehealth encounter (ref to EHR visit)
- Provider & patient mapping checks
- Charge lines add/edit (CPT/HCPCS, units, charge amount)

- Attestation by provider (signed date/time)
- Link documentation URI (notes, screenshots, consents)

Entities:

- **Encounter**
 - EncounterID
 - PatientID
 - ProviderID
 - EncounterDateTime
 - VisitType (Audio/Video/Async)
 - POS (02/10)
 - DocumentationURI
 - Status (Open/ReadyForCoding/Finalized)
- **ChargeLine**
 - ChargeID
 - EncounterID
 - CPT_HCPCS
 - Modifiers (JSON)
 - Units
 - ChargeAmount
 - RevenueCode (optional)
 - Notes
 - Status (Draft/Finalized)
- **Attestation**
 - AttestID
 - EncounterID
 - ProviderID
 - AttestText
 - AttestDate
 - Status

4.5 Coding, Modifiers & POS Module

Features:

- ICD-10-CM diagnosis coding (primary/secondary)
- Telehealth modifiers (e.g., 95, GT, FQ, 93 as applicable per policy)
- POS 02 (Telehealth other than patient's home) / POS 10 (Telehealth patient's home) selection
- CCI edits placeholder; payer-specific rules mapping
- Coding validation checklist & lock for claim

Entities:

- **Diagnosis**
 - DxID
 - EncounterID
 - ICD10Code
 - Description
 - Sequence (1..n)
 - Status
- **CodingLock**
 - CodingLockID
 - EncounterID
 - CoderID
 - LockedDate
 - Notes
 - Status (Locked/Unlocked)

4.6 Claim Building (837P Ref) & Scrubbing Rules Module**Features:**

- Claim assembly (header, subscriber, patient, provider, service lines)
- Use payer plan & fee schedule for allowed amounts
- Scrubbing rules (missing POS/modifiers/DX pointers/invalid NPI/date conflicts)
- Error queue with corrective actions
- Claim ready flag & batch assignment

Entities:

- **Claim**
 - ClaimID
 - EncounterID
 - PatientID
 - PlanID
 - SubscriberRel (Self/Spouse/Child/Other)
 - TotalCharge
 - ClaimStatus
(Draft/ScrubError/Ready/Batched/Submitted/Accepted/Rejected/Denied/Paid/PartiallyPaid)
 - CreatedDate
- **ClaimLine**
 - ClaimLineID
 - ClaimID
 - LineNo
 - CPT_HCPCS
 - Modifiers (JSON)
 - Units
 - ChargeAmount
 - DxPointers (JSON)
 - POS
 - LineStatus
- **ScrubRule**
 - RuleID
 - Name
 - ExpressionJSON
 - Severity (Error/Warning)
 - Status
- **ScrubIssue**
 - IssueID

- ClaimID
- ClaimLineID
- RuleID
- Message
- DetectedDate
- Status (Open/Resolved)
- **X12_837P_Ref** (*reference*)
 - X12ID
 - ClaimID
 - PayloadURI
 - GeneratedDate
 - Version
 - Status

4.7 Prior Authorization & Attachments Module (Reference)

Features:

- PA tracking by payer plan, linking encounter and claim
- Upload supporting documents (clinical notes, consents)
- PA status & expiry tracking; reminders

Entities:

- **PriorAuth**
 - PAID
 - ClaimID
 - PlanID
 - AuthNumber
 - RequestedDate
 - ApprovedFrom
 - ApprovedTo
 - Status (Requested/Approved/Denied/Expired)
- **AttachmentRef**
 - AttachID

- ClaimID
- FileType (PDF/Image/Doc)
- FileURI
- Notes
- UploadedBy
- UploadedDate
- Status

4.8 Submission, Clearinghouse & Payer ACK Module (Reference)

Features:

- Batch selection & submit (837P reference payload)
- Clearinghouse & payer acknowledgements (999/277CA references)
- Store ack status & correlation IDs
- Resubmit/corrected claim indicators

Entities:

- **SubmissionBatch**
 - BatchID
 - BatchDate
 - ItemCount
 - TotalCharge
 - Status (Open/Generated/Submitted/Acked/Failed)
- **SubmissionRef**
 - SubmitID
 - BatchID
 - ClaimID
 - ClearinghouseID
 - CorrelationID
 - SubmitDate
 - AckType (999/277CA/Other)
 - AckStatus (Accepted/Rejected)
 - AckDate

- Notes

4.9 ERA/EOB Posting, Patient Balance & Statements Module (Reference)

Features:

- ERA (835) payload reference & parse results into payments/adjustments (reference)
- Post allowed/paid/adjustment/write-off refs to claim lines
- Compute patient responsibility (copay/coinsurance/deductible)
- Statement generation & aging references

Entities:

- **RemitRef**
 - RemitID
 - PayerID
 - BatchID
 - PayloadURI
 - ReceivedDate
 - Status (Loaded/Posted/Failed)
- **PaymentPost**
 - PaymentID
 - ClaimID
 - ClaimLineID
 - AmountPaid
 - AdjustmentJSON
 - PostedDate
 - PostedBy
 - Status
- **PatientBalance**
 - BalanceID
 - PatientID
 - ClaimID
 - AmountDue

- AgingBucket (0-30/31-60/61-90/90+)
- LastStatementDate
- Status
- **Statement**
 - StatementID
 - PatientID
 - PeriodStart
 - PeriodEnd
 - GeneratedDate
 - AmountDue
 - SummaryJSON
 - Status (Open/Closed)

4.10 Denials, Appeals & AR Worklists Module

Features:

- Denial codes/reasons & mapping (CARC/RARC ref)
- AR worklists by bucket, payer, claim age
- Appeal tracking with letters & attachments (reference)
- Follow-up notes & next-action reminders

Entities:

- **Denial**
 - DenialID
 - ClaimID
 - ClaimLineID
 - ReasonCode
 - RemarkCode
 - DenialDate
 - AmountDenied
 - Status (Open/Appeal/Closed)
- **Appeal**
 - AppealID

- DenialID
- SubmittedDate
- Method (Portal/Mail/Fax)
- AttachmentURI
- Outcome (Pending/Won/Lost)
- OutcomeDate
- Notes
- **ARWorkitem**
 - WorkID
 - ClaimID
 - Priority (High/Normal)
 - AssignedTo
 - NextActionDate
 - Notes
 - Status (Open/InProgress/Done)

4.11 Reports, Dashboards & Audit Module

Features:

- KPIs: Clean claim rate, first-pass acceptance, DSO, denial rate, payer TAT
- Exports: Claims listing, scrub issues, AR aging, statements, remit summaries
- Immutable audit log search & export
- Scheduled report packs

Entities:

- **BillingReport**
 - ReportID
 - Scope (Payer/Plan/Provider/Period)
 - MetricsJSON (CCR/FPAR/DSO/DenialRate/TAT)
 - GeneratedDate

4.12 Notifications & Alerts Module

Features:

- Scrub errors & claim readiness alerts
- Batch submission & ack change alerts
- Remit received/posted refs
- Denial creation & appeal deadline alerts
- Statement generation & payment reminder alerts (in-app only)

Entities:

- **Notification**
 - NotificationID
 - UserID
 - Message
 - Category (Scrub/Submission/Ack/Remit/Denial/Statement)
 - Status (Unread/Read/Dismissed)
 - CreatedDate

5. Deployment Strategy

- **Local:** Angular/React UI, Spring Boot/.NET Core backend, local RDBMS
- **Production:**
 - API gateway/WAF; stateless microservices; container orchestration
 - Schedulers for **scrub batches, submission packs, ERA posting refs, AR worklists, statements**
 - Object storage for claim X12 refs, attachments, statements
 - Observability: central logs/metrics/traces; alerts on job failures & SLA breaches
 - Backups/DR: daily backups + PITR; environment promotion strategy; secrets via vault/KMS

6. Database Design

Tables:

- User(UserID, Name, Role, Email, Phone, Status)
- AuditLog(AuditID, UserID, Action, Resource, Timestamp, Metadata)
- Provider(ProviderID, NPI, Name, Taxonomy, TelehealthEnrolled, ContactInfo, Status)

- Payer(PayerID, Name, PayerCode, ClearinghouseCode, ContactInfo, Status)
- PayerPlan(PlanID, PayerID, PlanName, NetworkType, POSDefault, TelehealthModifiersJSON, Status)
- FeeSchedule(FeeID, PlanID, CPT_HCPCS, ModifierCombo, AllowedAmount, EffectiveFrom, EffectiveTo, Status)
- Patient(PatientID, MRN, Name, DOB, Gender, AddressJSON, ContactInfo, Status)
- Coverage(CoverageID, PatientID, PlanID, MemberID, GroupNumber, EffectiveFrom, EffectiveTo, Status)
- EligibilityRef(EligibilityID, CoverageID, RequestPayloadURI, ResponsePayloadURI, CheckedDate, Result, Notes)
- Encounter(EncounterID, PatientID, ProviderID, EncounterDateTime, VisitType, POS, DocumentationURI, Status)
- ChargeLine(ChargeID, EncounterID, CPT_HCPCS, Modifiers, Units, ChargeAmount, RevenueCode, Notes, Status)
- Attestation(AttestID, EncounterID, ProviderID, AttestText, AttestDate, Status)
- Diagnosis(DxID, EncounterID, ICD10Code, Description, Sequence, Status)
- CodingLock(CodingLockID, EncounterID, CoderID, LockedDate, Notes, Status)
- Claim(ClaimID, EncounterID, PatientID, PlanID, SubscriberRel, TotalCharge, ClaimStatus, CreatedDate)
- ClaimLine(ClaimLineID, ClaimID, LineNo, CPT_HCPCS, Modifiers, Units, ChargeAmount, DxPointers, POS, LineStatus)
- ScrubRule(RuleID, Name, ExpressionJSON, Severity, Status)
- ScrubIssue(IssueID, ClaimID, ClaimLineID, RuleID, Message, DetectedDate, Status)
- X12_837P_Ref(X12ID, ClaimID, PayloadURI, GeneratedDate, Version, Status)
- PriorAuth(PAID, ClaimID, PlanID, AuthNumber, RequestedDate, ApprovedFrom, ApprovedTo, Status)
- AttachmentRef(AttachID, ClaimID, FileType, FileURI, Notes, UploadedBy, UploadedDate, Status)
- SubmissionBatch(BatchID, BatchDate, ItemCount, TotalCharge, Status)
- SubmissionRef(SubmitID, BatchID, ClaimID, ClearinghouseID, CorrelationID, SubmitDate, AckType, AckStatus, AckDate, Notes)
- RemitRef(RemitID, PayerID, BatchID, PayloadURI, ReceivedDate, Status)
- PaymentPost(PaymentID, ClaimID, ClaimLineID, AmountPaid, AdjustmentJSON, PostedDate, PostedBy, Status)

- PatientBalance(BalanceID, PatientID, ClaimID, AmountDue, AgingBucket, LastStatementDate, Status)
- Statement(StatementID, PatientID, PeriodStart, PeriodEnd, GeneratedDate, AmountDue, SummaryJSON, Status)
- Denial(DenialID, ClaimID, ClaimLineID, ReasonCode, RemarkCode, DenialDate, AmountDenied, Status)
- Appeal(AppealID, DenialID, SubmittedDate, Method, AttachmentURI, Outcome, OutcomeDate, Notes)
- ARWorkitem(WorkID, ClaimID, Priority, AssignedTo, NextActionDate, Notes, Status)
- BillingReport(ReportID, Scope, MetricsJSON, GeneratedDate)
- Notification(NotificationID, UserID, Message, Category, Status, CreatedDate)

7. User Interface Design

- **Charge Capture & Coding:** Encounter list, charge lines, DX entry, POS/modifiers, coding lock, checklist
- **Claim Workbench:** Scrub results, fix console, line-level issues, ready flag, batch assignment
- **Submission Desk:** Batch builder, X12 preview (ref), submit log, acknowledgements (ref)
- **AR & Denials:** ERA/EOB refs, posting view, denials buckets, appeal tracker, worklists
- **Statements & Patient Balance:** Statement generator, aging dashboard, download refs
- **Master Data:** Providers, payers/plans, fee schedules, rule editor
- **Reports & Audit:** KPIs, exports (CSV/PDF), audit log search
- **Admin:** Users/roles, sites, parameters, schedulers, feature flags

8. Non-Functional Requirements

- **Performance:** P95 < 2s for claim workbench loads; scrub batch for 5k claims < 10 min; statement run 10k patients < 20 min
- **Security:** RBAC, TLS, encryption at rest (config); immutable audit; PHI minimization; session hardening; least-privilege
- **Scalability:** Horizontally scalable stateless services; async workers for scrub/submit/statement/AR packs; partition by payer/plan

- **Availability: 99.9% uptime**; graceful degradation (defer heavy reports during peak)
- **Maintainability**: Modular services; configuration-driven rules & fee schedules; API versioning; automated DB migrations
- **Observability**: Central logs, metrics, traces; KPIs (clean claim rate, FPAR, DSO, denial rate); alerting on job failures/spikes
- **Data Integrity**: Transactional claim assembly; idempotent submission & posting refs; full audit chain

9. Assumptions & Constraints

- **Phase-1** excludes live clearinghouse/payer integrations—**X12/ACK/835 stored as references** only
- e-eligibility and prior auth are **references**; results recorded but not transacted
- No patient pay collection gateway in scope; statements are informational references
- In-app notifications only; no SMS/email in Phase-1
- Fully implementable with **Java Spring Boot/ASP.NET Core + Angular/React + RDBMS**